enables to calculate the probability even without conducting the experiment but using the prior knowledge about the experiment, it is also called as a priori probability.

Example 8.5

What is the chance of getting a king in a draw from a pack of 52 cards?

Solution:

In a pack there are 52 cards [n(s) = 52] which is shown in fig. 8.2

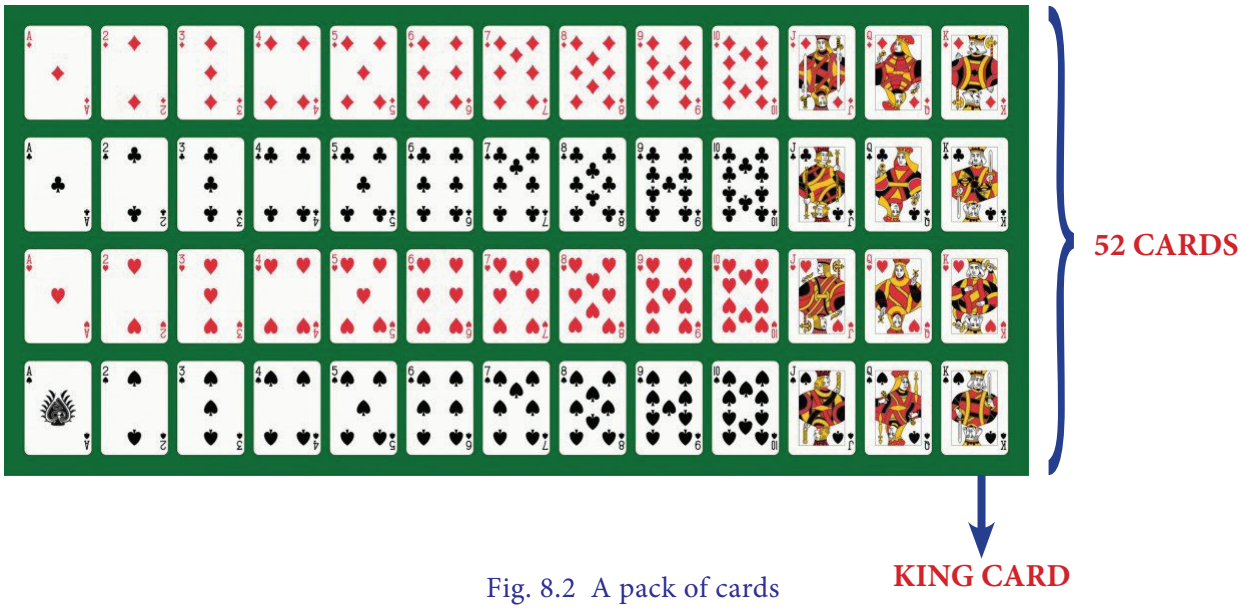


Fig. 8.2 A pack of cards

Let A be the event of choosing a card which is a king

In which, number of king cards $n(A) = 4$

Therefore probability of drawing a card which is king is $= P(A) = \frac{n(A)}{n(S)} = \frac{4}{52}$

Example 8.6

A bag contains 7 red, 12 blue and 4 green balls. What is the probability that 3 balls drawn are all blue?

Solution:

Out of 23 balls 3 balls can be chosen $23C_3$ ways

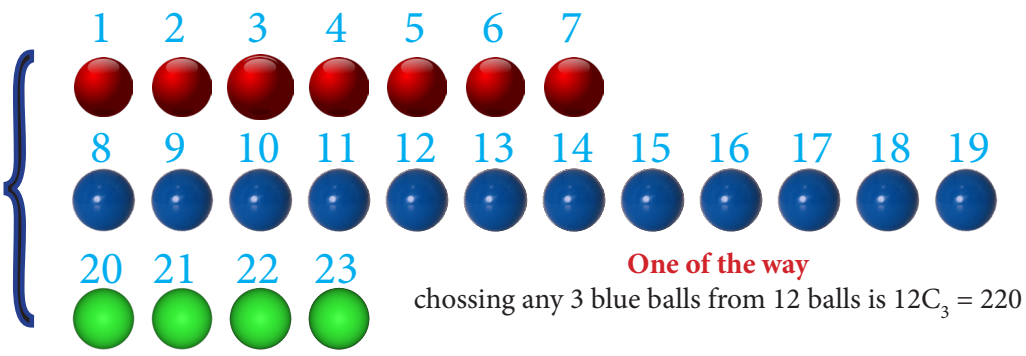


Fig. 8.3 Selection of 3 blue balls out of 23 balls